

Gania Rafidah Huwaida

Computer Engineering student at Universitas Brawijaya with a 3.65/4.00 GPA. During a six-month BRIN research internship, I built connected systems for EV charging analytics, OCPP simulation, and real-time industrial energy monitoring.



Computer Engineering - AI, IoT, Data

6+

Projects Built

2+

National Internships

3.65

GPA / 4.00

70+

Staff Led

ABOUT

Building tools people can *use*.

I am drawn to systems where software touches the physical world: a charger exchanging protocol messages, a smart meter streaming imperfect field data, a microphone turning speech into action, or a mobile camera running a model offline. From February to July 2026, I joined BRIN's E-Mobility research group and delivered Enerlytics, ChargeSim, and an EMS Modbus pipeline. Earlier at Telkom Indonesia BigBox, I explored local speech-to-speech AI with DSP, ASR, FastAPI, and CPU-friendly model choices.

WORK EXPERIENCE

Dec 2025 - Jan 2026

Data & Automation Intern - Telkom Indonesia BigBox

- Built a local speech-to-speech assistant using Python, FastAPI, Faster-Whisper, PyAudio, FFmpeg, VAD, and gTTS.
- Designed CPU-friendly int8 inference and RAW vs CLEAN audio comparison, then set up a small n8n, Docker, and Google Sheets API retrieval workflow.

Feb 2026 - Jul 2026

Machine Learning Research Intern - BRIN

- Delivered three connected prototypes during a six-month placement with the E-Mobility research group.
- Built Enerlytics for multi-source energy analytics and seven-model forecasting, and ChargeSim for OCPP 1.6 testing, queue management, and smart power sharing.
- Built an EMS Modbus pipeline for three smart meters with MySQL logging, MQTT distribution, and secured Streamlit monitoring.

Jun 2023 - Jul 2023

Team Leader, Operations - Jakarta Fair

- Led around 70 operational staff during a 30-day national-scale event.
- Managed attendance, shift allocation, field coordination, handoffs, and escalation during high-traffic operations.

SKILLS

Backend & Data

Python, Flask, FastAPI, MySQL, SQLAlchemy, SQL, Docker, REST API, ETL, CSV/XLSX processing, dashboard data flow.

Machine Learning & AI

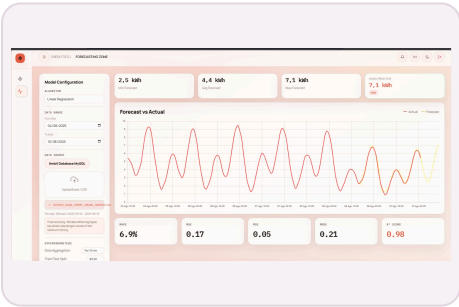
scikit-learn, XGBoost, CatBoost, LSTM, TensorFlow/Keras, PyTorch, MobileNetV4, transfer learning, forecasting, classification.

Audio, IoT & Frontend

OCPP 1.6-J, WebSocket, Modbus TCP, MQTT, asyncio, PySide6, Streamlit, Raspberry Pi, Faster-Whisper, FFmpeg, Chart.js, Android, Kotlin.

SELECTED PROJECTS

Projects with *working context*.



EV charging forecasting dashboard

Enerlytics - Multi-Source Energy Analytics & Forecasting

A Flask platform that unifies OCPP charger data, Modbus meters, MQTT telemetry, file uploads, and API input through 20+ REST endpoints and a 12+ table SQLAlchemy schema.

Implemented seven forecasting approaches with ordered weekly evaluation, comprehensive metrics, automatic interpretation, and safeguards against implausible future predictions.

Python

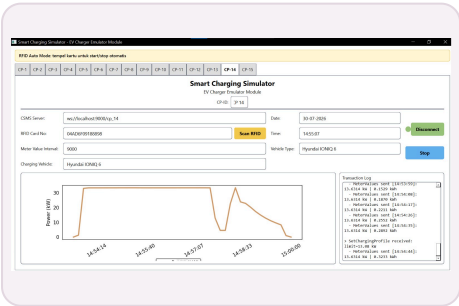
Flask

OCPP 1.6

Modbus TCP

MySQL

Forecasting



OCPP charge point simulator

ChargeSim - OCPP Smart Charging Simulator

A Python and PySide6 CSMS testbed with up to 15 charge-point slots, queue management, remote operations, live power views, smart power sharing, MySQL transaction storage, CSV export, and Raspberry Pi support.

Validated basic connectivity with a BRIN physical charger through WebSocket, StatusNotification, and Heartbeat; full physical transactions and charging-profile application remain follow-up work. Repository: github.com/gnieerfd/smart-charging-simulator.

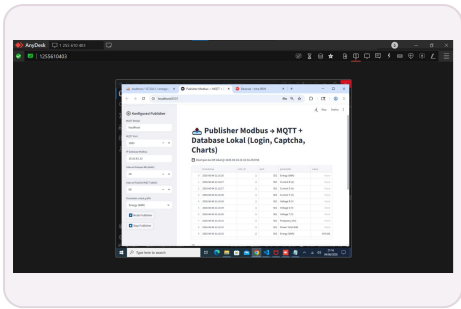
OCPP 1.6-J

PySide6

WebSocket

MySQL

Raspberry Pi



Operational Modbus, MQTT, and local database dashboard

EMS Modbus - Industrial Energy Monitoring

A real-time pipeline for nine energy parameters across three smart meters, with raw-register validation, endianness and scaling handling, bounded buffering, MySQL time-series logging, and MQTT publishing.

The secured Streamlit dashboard adds bcrypt password hashing, role-aware access, arithmetic CAPTCHA, live tables, and trend charts.

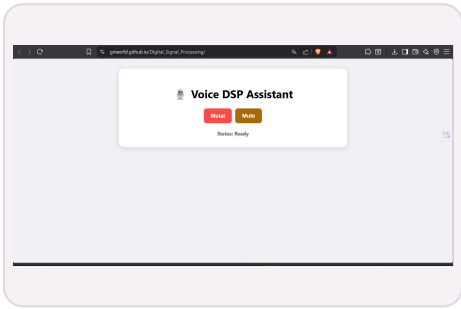
Modbus TCP

MQTT

Streamlit

SQLAlchemy

MySQL



Local speech-to-speech assistant

Jarvis S2S - Local Voice AI

A local speech-to-speech assistant with microphone capture, custom VAD, Faster-Whisper ASR, LLM response handling, TTS output, and a FastAPI upload mode.

Uses int8 CPU inference and FFmpeg noise reduction for privacy-friendly speech AI on modest hardware. Repository: github.com/gnieerfd/Local_Speech-to-Speech_AI-Assistant.

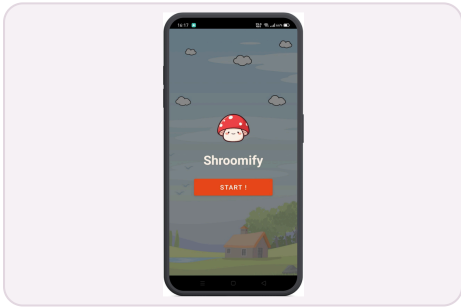
Faster-Whisper

FastAPI

PyAudio

FFmpeg

Docker



Offline mushroom classification

Shroomify - Offline Mushroom Classifier

Owned the ML workflow for an Android mushroom classifier: dataset cleanup, augmentation, MobileNetV4 training, fine-tuning, and model export.

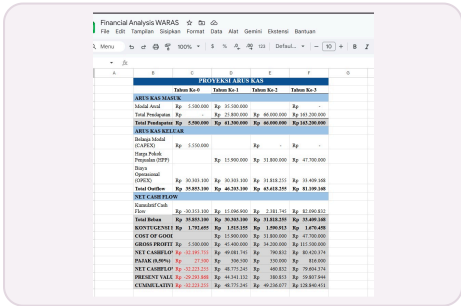
Focused on binary edible/poisonous classification for offline mobile inference using TorchScript and PyTorch Mobile.

MobileNetV4

TensorFlow/Keras

PyTorch Mobile

Transfer Learning



Business plan and financial model

WARAS - AI-Based Cyber Safety Business Plan

An AI-based plug-and-play micro-gateway concept using DNS/SNI filtering, Random Forest classification, selective redirection, Orange Pi Zero 3 hardware, and custom Linux firmware concepts.

Mapped Rp255,000,000 projected 3-year revenue, 101-unit BEP, 70.98% ROI, 108.75% IRR, and 0.34-year payback. Finalist at PRISMA 2025.

Random Forest

DNS/SNI

Orange Pi

Financial Analysis

ORGANISATION & LEADERSHIP

BEM FILKOM UB 2024

Core Team for Leader of Tomorrow 6.0 and Pengguna Anggaran for Leader of Tomorrow Chapter 2. Supported attendance records, PUMK documents, LPJ flow, funding files, and faculty-level reporting. Recognized as Best Staff - November.

Raja Brawijaya 2025

Processed and checked 800+ participant records for attendance, QR usage, validation, and event reporting, reducing friction during large-scale attendance checks.

Global Liaison Officer

Supported Singapore Polytechnic and Vietnam delegation visits through guest guidance, communication flow, campus tours, and schedule movement.

Dev Girls & KarsaCita

Supported UI/UX workshops, mentoring sessions, partner coordination with Skilvul, and online sessions with 50+ participants on average.

ACHIEVEMENTS

Finalist

WARAS Business Plan Competition, PRISMA UINSA 2025.

Best Staff

BEM FILKOM UB 2024, November recognition.

Academic Awards

Gold Medal - Computer Science, Bronze Medal - Mathematics, and Microsoft Azure AI Fundamentals.

CONTACT

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